Vuk Topalovic

Senior Full-Stack Engineer

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PROFILE

Full-stack engineer who builds web platforms end to end, from the data model and API to the interface. Since 2022 I have worked on EU-funded platforms for energy, buildings, smart cities and secure telemetry at European Dynamics, in small remote teams where I own features from requirements to deployment. I also build and publish developer tools, and I design them to measure their own results.

SKILLS

Frontend	TypeScript, React, Next.js, Angular, React Query, Jest, Testing Library
Backend	Java, Spring Boot, Spring Security, Hibernate, Node.js, Express, MQTT
Data & infrastructure	PostgreSQL, MongoDB, MySQL, Prisma, Docker, Jenkins, GitHub Actions, Linux
Security	Keycloak, OAuth2, JWT, RBAC
AI tooling	Anthropic SDK, OpenAI SDK, Vercel AI SDK, pgvector, Zod

EXPERIENCE

Full-Stack Software Engineer · European Dynamics — Dec 2022 — Present · Remote (Athens, Greece)

Full-stack work on EU-funded research and innovation platforms, in remote teams of four to five engineers.

- **OpenBEP4EU.** Full-stack development for an EU LIFE project that turns the ISO 52000 energy-performance standards into open calculation methods for Energy Performance Certificates and the Smart Readiness Indicator.
- **EnerShare.** Designed and shipped secure REST APIs for a European energy data space aligned with IDSA and GAIA-X, supporting 7 pilot deployments and 11 cross-sector use cases.
- **Resonance.** Built real-time MQTT pipelines that bring sensor data from devices in several cities into one backend, plus marketplace features and backend services.
- **UnderSec.** Implemented authentication across microservices with Keycloak, JWT and RBAC; containerised the services with Docker and ran CI/CD on Jenkins and GitHub Actions.
- **Tenacity RMT.** Built a React and TypeScript risk-management dashboard with dynamic filtering, charts and reporting, integrated with existing Angular applications.

Java · Spring Boot · Node.js · TypeScript · React · PostgreSQL · Keycloak · MQTT · Docker

Software Engineer · Freelance — 2026 — Present · Remote

Automation and backend work for independent clients.

- **Port appointment booking.** Built a TypeScript and Node.js service for a logistics client that books truck appointments at Saudi land ports through the FASAH platform for a whole team of drivers.
- **Slot ranking.** Orders open slots by each driver's preferred hours, then assigns drivers to distinct slots with min-cost matching (Hungarian algorithm), so they never compete for the same one.
- **Reliability.** Adaptive concurrency that drops to one request at a time under rate limits, per-token budgets, scheduled search windows and detailed logs. Now porting the core to Rust.

TypeScript · Node.js · Rust

Operations Coordinator · Robert Bosch — Mar 2019 — Dec 2022 · Belgrade, Serbia

SELECTED PROJECTS

token-warden

github.com/vukkt/token-warden

A Claude Code plugin that makes coding agents cheaper over time.

Agent memory is usually a file of advice that nobody measures. token-warden records what each session costs, drafts candidate rules from the expensive ones, and benchmarks every rule on a frozen test suite. A rule stays only if it saves at least twice what it costs to carry.

TypeScript · Node.js · SQLite · Zod · Claude Code

iglass

github.com/vukkt/iglass

A Liquid Glass theme for iTerm2 and Ghostty.

Brings the macOS Liquid Glass look to the terminal: translucent windows, Apple's system colors and four glass levels from Clear to Solid. Every color is contrast-checked against the window as it looks over the desktop, and uninstalling restores your setup exactly.

Shell · Python · jq · Homebrew

PR Scout

npmjs.com/package/pr-scout

Finds beginner-friendly GitHub issues worth your time.

Fetches open good-first-issue tickets from any repository and has an LLM score each one for clarity, scope, spam signals and actionability. It skips stale projects by checking stars, commit recency and merged pull requests, then lets you open the best issues from an interactive list.

TypeScript · Node.js · GitHub API · Groq

Trinomen

npmjs.com/package/trinomen

A three-agent LLM pipeline in the terminal: route, generate, review.

A router classifies the request, a worker generates code and a reviewer on a second provider audits it. Benchmarks showed reviewing every output cost about five times the tokens for worse results, so the loop was redesigned to typecheck first.

Node.js · Vercel AI SDK · Zod · SQLite

EDUCATION

BSc, Road Traffic, Faculty of Transport and Traffic Engineering, University of Belgrade — 2014 — 2019

Java Spring Boot Master Class, Amigoscode — 2022

LANGUAGES

Serbian (native), English (professional working proficiency)